Results

These results are primarily from about 15 microatolls from Ranonnga Island to Vella Mbava Island in the West and then eastward to Ghizo and smaller islands and onward to the western tip of Parara Island (Fig.). (Fig.). 12 MBU-A and 12MBU-B lie on or very near the 2007 hingeline between uplift and subsidence ~35 km arcward of the trench. Most microatoll sites lie on the arcward portion of the coseismic uplift area and overlie the April 2007 megathrust rupture zone inferred from the vertical deformation pattern (e.g., Taylor et al., 2008).

Other microatoll samples and observations are located southeast of the 2007 rupture zone on Rendova and Tetepare Islands. Microatoll 12UGH-A is from a site on central Rendova and microatolls 12RAR-A and 12TET-A are from sites along the north coast of Tetepare. A major arc segment boundary is confirmed by the 2007 rupture zone along a line paralleling the linear northwestern coast of Rendova (e.g., Chen et al., 2012). These sites are important in part because they must share regional sea-level with microatolls that were displaced by the 2007 event, but are unlikely to share the same vertical motions. However, the Rendova and Tetepare corals are on the part of the forearc most prone to vertical tectonism so that we may expect they have undergone some vertical tectonic motions.

To reinforce the microatoll sample data we have added relevant observations from other living and dead microatolls. A few of these observations are from the inner forearc that subsided in association with the April 2007 megathrust rupture. The coral response to subsidence is upgrowth to take advantage of increased accommodation space. But even with zero vertical tectonics, many corals may have already been growing upward in response to sea level rise.

To provide further context, we refer to Holocene uplift rates from previous and current results and the dates of the most recent paleouplifts . The net result is not a comprehensive paleouplift survey mapping every location, but rather a more generalized accounting of vertical deformation of the century or so and the revelation of abundant variability in timing and amounts of vertical motion. To fully map the timing and amounts of vertical motions and the geographic limits of the numerous blocks over an area exceeding 10,000 km^2 would require higher density sampling.

Ranongga Island northward to Mbava Island

Some of the microatoll sites on Ranongga and islets to the north are also sites where mean Holocene uplift rates and several of the most recent paleouplifts serve as context for the microatoll analyses (Mann et al., 1998; Thirumalai et al., 2015). Holocene uplift rates exceed 5 mm/yr on southeastern Ranongga (Konggu area) where the Holocene reef reaches ~35 m ALC (Mann et al., 1998; Taylor et al., 2008; Thirumalai et al., 2015). These uplift rates decrease northward to zero net Holocene uplift at Ghoi and Nakassa Islands off the northern tip of Ranongga. Holocene uplift increases again at Mbava Island and along the southern and western coasts of Vella Lavella.

Lale Village Area, Southernmost Ranongga Island

Lale Village near the southern tip of Ranongga Island, uplifted ~ 2.46 m in 2007 (Taylor et al., 2008; others 200x). However, total uplift since the mid-Holocene was only ~11 m at the southern end of the island counting the 2007 uplift. Interviews with the Angelo family who live there, in particular Florida Angelo, revealed that this area had been subsiding rapidly in the years before the 2007 coseismic uplift. Florida's father, John Angelo, stated that in his youth, about 60 years ago when the area of thriving coral reef uplifted in 2007 had been dry land. This is consistent with our 1991 field observations of severe coastal erosion with coconut trees being undermined and foundering. We interpreted this as indicating ongoing subsidence and marine transgression.

Porites lutea coral heads up to at least 1.7 m tall had grown upward until 2007 on formerly dry land that was inundated by the encroaching sea as Ranongga subsided (Fig. x). Porites lutea samples show that rapid upward coral growth has proceeded for at least 140 yrs with no recognizable pauses in upward growth that could be due to even temporary emergence of living coral surfaces. This, combined with witnesses to century-scale subsidence, indicates that rising RSL outpaced coral upgrowth, which is typically ~10-14 mm/yr in these corals. Considering that the witnesses stated that the shoreline was many meters seaward of the corals we sampled, it is likely that total subsidence exceeded 1.7 m and may have equaled the 2.46 m of 2007 coseismic uplift.

Lale corals had resumed submerging in 2012 and 2013 when we last visited. The amount of upgrowth exceeded 6 cm on some corals. This strongly suggests that coral upgrowth in response to tectonic subsidence resumed soon after the coral heads had been partially killed by 2007 coseismic uplift. Total re-submergence was at least 6 cm by August 2013. GPS measurements confirm ongoing subsidence on the order of 10 mm/yr (Kuo et al., 2016).

Lale Interpretation

Total Holocene tectonic emergence of ~11 m (counting 2.46 m in 2007) indicates net Holocene uplift of ~9 m at a mean rate of ~ 1 mm/yr. If this area was raised by a series of permanent 2.5 m uplifts the only 3.6 uplifts would raise it by 9 m at a mean interval of ~ 1667 yrs. However, the large amount of subsidence preceding the 2007 event suggests that only a small amount of uplift may have resulted from this most recent earthquake cycle. Six ^{230}Th dates on corals from 4.84 to 9.46 m ALC at Lale range in age from 5960 to 6581 years (T et al., 2015). Our unsuccessful search for paleoseismically uplifted corals in the 5 to 0 ka range is not conclusive, but the apparent absence of evidence for younger corals, the large subsidence preceding the 2007 uplift, and resumption of subsidence soon after 2007 suggests that subsidence has erased evidence of most late

Holocene coseismic uplifts by exposing the uplifted corals to bioerosion as they subside down through the intertidal zone. In contrast, only ~7 km up the coast at Konggu, where net Holocene uplift exceeds 35 m, there is evidence of several permanent late Holocene uplifts.

Konggu

Kongu Village is only 6-7 km northeast of Lale, but rather than the 11 m of Holocene emergence at Lale, in 2012 at Kongu the mid-Holocene reef forms a terrace surface at 35.84 m above living coral. Mann et al. (1998) dated a large in situ *D. heliopora* coral head at ~35.5 m ALC from the top of the seaward scarp below the terrace at 7025± 60 yr BP (uncorr. ^{14}C date, $t^{1/2}$ 5568 yr). This indicates a mean uplift rate on the order of ~5.5 mm/yr. Thirumalai et al. (2015) identified and dated the most recent uplifted paleoshorelines at 614.3± 2.7 yr and one that gave ages of 812.1 ±8.6, 822.2 ±3.4 yr An additional coral from this level was rather eroded and so gave an older age of 846.6 ± 3.0. The difference between Konggu and Lale is that Lale underwent large amounts of pre-2007 subsidence. Konggu subsided relatively little prior to 2007 so that it may retain more permanent uplift (Thirumalai et al., 2015).

At Konggu, 2007 uplift raised corals by up to 2.6 m exposing them for close inspection. Most of the higher corals killed by 2007 coseismic uplift were relatively small. Many had a dead top about 30-40 cm below the upper part of a collar of coral that had grown upward from the living periphery of the corals. The dead surfaces of these corals were extremely fresh because they were abruptly uplifted well above the highest HTLs. However, in one location we found a cluster of small coral heads that appeared to be clearly older and higher then those with much better preserved surfaces. These corals gave ^{230}Th ages that ranged from 13.3 ± 1.5 to 26.96 ± 0.87 yr before 2013. After living at that level for on the order of a few decades these corals emerged enough that they were killed. Their deaths could have been facilitated of ENSO-related lower sea levels. However, their demises also suggest that subsidence that had previously been providing accommodation space had ceased so that they were more vulnerable. It also is possible that there was some tectonic uplift that raised these higher corals to depths too shallow for longer-term survival.

To summarize, the somewhat larger and lower microatolls record an emergence event that killed their tops in approximately 1970. We do not know if there was submergence prior to ~1970 or if the corals had simply colonized a substrate and were growing upward until their tops were killed. Tectonic uplift could have caused the microatoll tops to die by emergence. Either abrupt or gradual submergence of at least tens of cm followed so that the microatolls then grew upward by at least 40 cm without pause until they were abruptly uplifted well above high tide levels in 2007. Meanwhile, corals colonized the leading edge of the transgressing shoreline. From 1982 until ~2000 C.E., these higher corals died due to emergence. They are somewhat more weathered than those killed in 2007 because they remained just above the HLS, but under water during high tides. Meanwhile, the deeper microatolls continued growing upward because they started from deeper water after their tops were killed in about 1970. Prior to 1970 we have no evidence of either subsidence or uplifts and we have no evidence

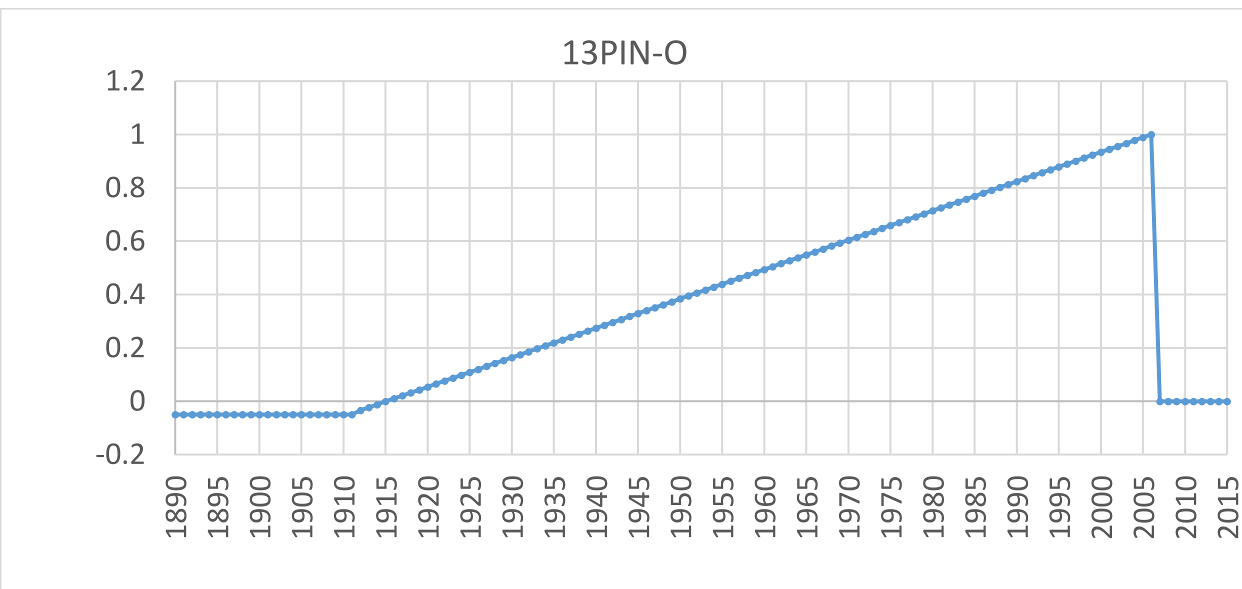
These corals as a group suggest a complex sequence of submer

Contrast these photos. At the top are corals killed by uplift emergence in April 2007 and the Below are corals that are a few decimeters higher and that give 230^{Th} ages for 12KON-A gave an age of 23.1 ± 1.5 (~1990 C.E.) and 12KON-C of 13.7 ±1.0 (~2000 C.E.) Nearby 12KON-B gave an age of 26.96 ± 0.9 (~1986 C.E.). For comparison, a coral that was certainly killed by the 2007 uplift gave a surface age of 10.2 ± 1.2

Southern Pienuna

South of Pienuna Village we found a 5-m diameter microatoll that was killed in 2007 by uplift that we measured as 2.15 m in 2013. During the earthquake the microatoll broke in half and is thus no longer properly oriented. However, we can see that it had a flat top with a living periphery. The living periphery had abruptly began growing upward and inward. By 2007, it had grown upward by measured heights of 1.00, 1.00 and 1.10 m. We dated a sample of the upgrowing coral periphery about 15 cm after the growth began (0.85-0.95 below top of coral) that gave an age of **88.7 ± 1.6 yr before April 2014, or 1926 CE. From 1926 until 2007 the coral then grew** 0.85-0.95 m at a mean rate of 11.1 mm/yr. Using this growth rate to correct for the 0.15 m of growth between initial upgrowth and the location of the dated sample adds 13.5 yr to the age of initial upgrowth or 1913 C.E. The floor of the central area of the coral is too rugged with knobby overgrowths for us to discern a relative sea-level history for that part.

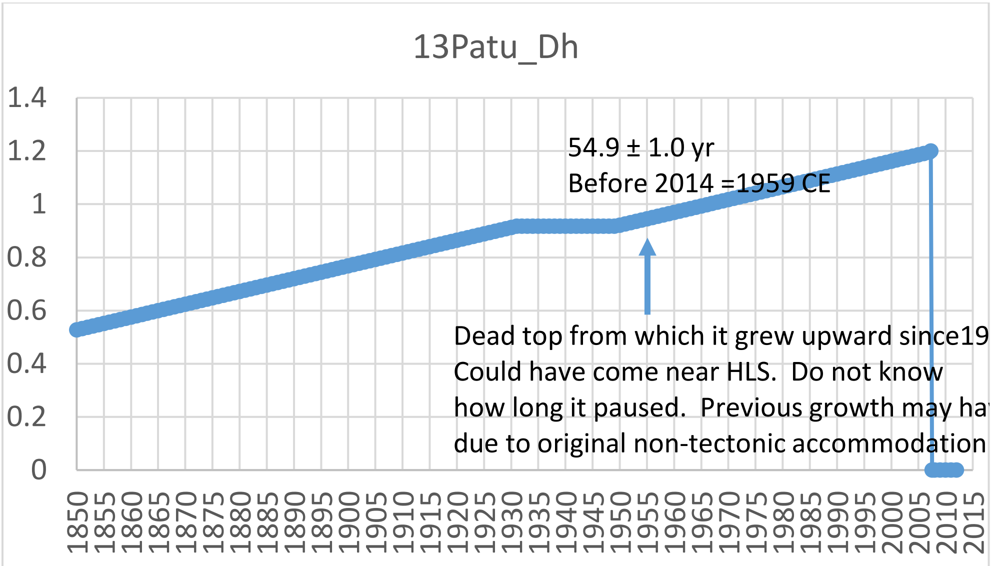
Alone, this coral indicates a minimum uplift of 1.0 to 1.1 m, however it was completely killed and uplift measured from nearby living and emerged 2007 corals is 2.15 m. The mean upgrowth rate of this coral was 11.1 mm/yr, which is consistent with rates from other corals in the area. Howwever, submergence of this coral may have exceeded its ability to keep up with relative sea level rise.



13 PATU-A

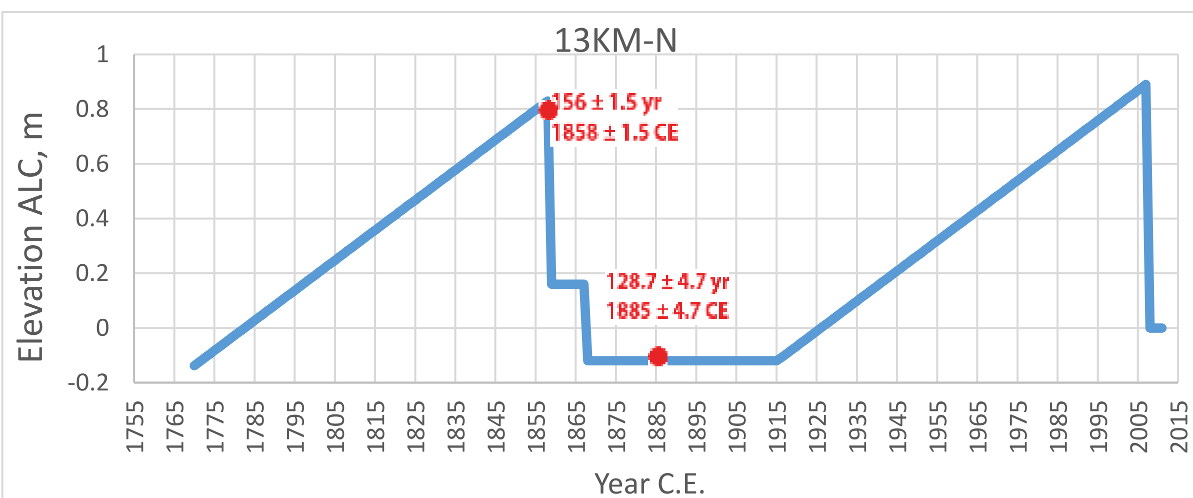
Farther north along the east coast of Ranongga we found two Diploastrea heliopora microatolls whose living tops were exposed during low tides. The tops of these colonies were killed by apparent emergence at some time in the past century or so. However, the top of the coral appears to have not been growing sideways, but rather upward when their tops were killed. Thus, the dead top may represent a brief pause in upgrowth rather than a period of stable relative sea level that lasted for decades. The sides of the coal had grew upward by ~30 cm, but it was well below current HLS until the 2007 uplift of ~1.57 m which was enough to kill the entire heads. Because this is a very slow-growing species (typically a few mm/yr), to have been exposed to water levels low enough to have killed its top requires that it had grown upward to near its HLS. This requires that relative sea level was not rising much as it grew upward prior to its top dying. It then subsided on the order of at least 0.5 m after its top died based on other corals from the area. A sample from 5 cm after upgrowth began gave an age of only 54.9 ± 1.0 yr before 2014 or 1959 C.E. Given a typical upgrowth rate of 3-4 mm/yr, the initial upgrowth would have begun ~12-17 yrs earlier or 1942 to 1947 C.E. Because this coral grows so slowly, it seems unlikely that it kept up with subsidence or that its top would have been killed by an ENSO low sea level. But this time of initial upgrowth is later than in corals to the north and south.

We represent this coral as having grown upward at a few mm/yr before about 1930 after which its top was killed. By about 1950, it resumed upgrowth of ~ 0.3 m and was killed by uplift in 2007. The role of relative sea level rise in upgrowth before 1940 is unknown. Other corals indicate that subsidence probably was occurring, but they are farther north and south on Ranongga and may have different tectonics.

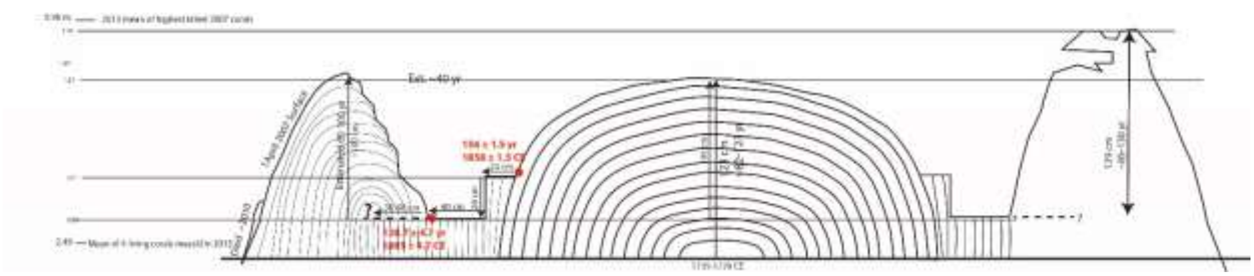


13 Kolomali North

At Kolomali the maximum Holocene reef elevation is ~18 m, which suggests a mean uplift rate of ~ 2.7 mm/yr or 2.7 m of permanent uplift every 1,000 yr.. The most recent permanent paleouplift was dated on two morphologically intact corals as 585.7 ± 2.5 and 603.5 ± 2.4 before 2012 or ~ 1408-1426 C.E. This was a single uplift in ~ 1417 C.E. These corals were almost perfectly preserved and were growing at their HLS limit 1.65 m higher than living 1991 living corals. In April 2007 this part of Ranongga uplifted ~1.6 m, which is a very similar amount, so that these fossil corals are now ~3.25 m ALC.



Immediately south of the emerged corals is a 4.75 m diameter *Porites lutea* microatoll located in shallow water at the north end of Kolomali Village beach. Two ^{230}Th ages of 156 ± 1.5 yr and 128.7 ± 4.7 yr before 2014 combined with coral morphology constrain the chronology. This microatoll shows not only the 2007 emergence, but a previous large emergence event, a period of relative sea- level stability, and then an initiation of coral upgrowth of ~1 m that was terminated by the 2007 coseismic uplift of ~1.6 m measured using living and 2007 corals located within tens of meters of this microatoll. It is notable that all of the 1858 ± 1.5 C.E. upgrowth of the coral was re-submerged by relative sea level rise that began abruptly in ~1915 C.E. It is possible that, as with the 2007 corals, that this microatoll upgrowth was lagging behind relative sea level rise and that the 1858 emergence was somewhat greater than ~1.0 m. After its large initial uplift in 1858 ± 1.5 C.E. the coral grew horizontally for 20 cm, which would be 18 yr at a typical growth rate of 11 mm/yr. It then emerged another 29 cm. From that point onward it again grew horizontally for at least 40 cm at which point subsequent coral growth hides the coral surface. It is notable that during the periods of horizontal growth there is no detectable upward or downward trend in the horizontal growth. The date of 1885 ± 4.7 C.E. is from just at the outer edge of exposed coral where subsequent growth began to cover it. The upgrowth relative to the horizontal surface was about 1.00 m on this side of the coral and 1.29 m on the other side. At 11 mm/yr, we estimate that the upgrowth began in about 91 to 117 yr ago. If we subtract these ages from 2007 then we estimate the upgrowth began between 1890 and 1916 C.E. . The 1890 C.E. age seems a little old given the location of the 1885 C.E. age. The 1916 C.E. age seems more consistent with allowing the coral to grow horizontally farther out before beginning to grow upward and inward over the horizontal dead surface of the coral. Unfortunately the coral representing initial upgrowth is covered so that we could not sample it.



The age control from two dates and using a typical growth rate of 11 mm/yr provide a consistent, but imperfect chronology for the coral. The 1885 age is only 27 yrs apart, but it appears that the coral grew horizontally by ~60 cm over this time. This is not likely. However the ~1 m of upgrowth of the coral preceding the 2007 earthquake is reasonably consistent with the ~1885 C.E. age for the last coral sample dated before the coral began growing upward. Until additional dates become available we choose to allow the dates to rule the chronology despite the unlikely growth rate required by the 27 yr difference in the two dates.

Microatoll 13KM-N began growing upward from the sea floor ~1770 C.E. It then uplifted on the order of 0.7 m in ~1858 ± 1.5 C.E. and grew horizontally in very stable relative sea level for ~18 yrs (using 11mm/yr growth rate). It then uplifted again by ~29 cm in ~1867 C.E. and continued growing horizontally for a few decades, again with no detectable relative sea level changes. In approximately 1915 C.E. it began to grow upward, and inward over the previously dead coral surface, by ~1 m. We could not see the initial upgrowth of the coral to tell if it began abruptly or not. In 2007 earthquake raised the coast by ~1.6 m and killed the coral. The upgrowth of >1 m is primarily due to subsidence, but may include some regional sea level rise, particularly the climatically driven sea level rise that occurred after 1993 C.E.

The 1417 C.E. and 1858 C.E. uplift events were very different because the older one produced ~1.65 m of permanent uplift and the 1858 C.E. uplift was completely erased from the coral record by subsequent relative sea level rise of which a large proportion was due to tectonic subsidence. We will have to wait to learn whether the 2007 uplift will be permanent or will be removed by subsidence. We do not know if subsidence preceded the 1417 C.E. uplift. If so, then the 1417 C.E. uplift could have been much larger than 1.65 m of retained permanent uplift. Or the 1417 C.E. uplift could have been more like the 2007 uplift at Konggu where very little subsidence preceded the earthquake so that most of the 2007 uplift may be permanent. (if it is the vm preceding the EQ that counts in the equation vs. the subsidence after the EQ).

Ghoi Island

Ghoi Island has undergone no Holocene uplift and may even have subsided based on the absence of emerged Holocene reef (Mann et al., 1998). In 2007, 3 km south of Ghoi the northern tip of Ranongga uplifted 0.98 m. We measured emergence of the largest microatoll which was nearly 11 m in diameter at its base. The living band of coral that survived 2007 uplift was growing outward, but not upward when examined in 2013. The flat floor of the microatoll averaged 0.99 m below the rim and the rim was a mean of 1.09 m higher than the living base. This represents a minimum uplift amount and suggests that the shallowest coral was below HLS water level. None of the microatolls in the area had reached HLS, so that relative sea level rise probably exceeded the typical upward growth rate of ~1 cm/yr. Therefore, total uplift may have been more than 1.09 m. Coral overgrowth obscured the central floor of the microatoll so that any relative sea level changes recorded there were unavailable.



The morphology of the microatolls around Ghoi Island indicate that they had lived near their HLS for at least a century prior to the onset of relative sea level rise that we attribute primarily to tectonic subsidence. Had there been a previous uplift, then it may have left a central hump in the coral as at KM-N. As best we can see, the central area of the microatoll is flat although it is obscured by overgrowth of mainly branching corals and debris. Coral upgrowth until 2007 was caused by relative sea level rise, mainly due to subsidence that provided accommodation space of at least 0.98 m. This would require approximately 100 years based on typical Porites spp. microatoll upgrowth rates. Thus we infer that subsidence may have begun in the time range 1913 to 1923 C.E. It also appears that the 2007 uplift raised the coral higher than the preceding HLS if our measurement was from solid microatoll floor material. We also note that the absence on ongoing subsidence in August 2013 is consistent with the flat floor of the microatoll that indicates growth over some period without subsidence. However, it also is possible that the coral was growing upward and its top was killed by a negative relative sea level excursion that could have been due to either or both tectonic or regional sea level. Obviously it would be useful to have a cross section of the microatoll.

Nakassa Island

There is no evidence of Holocene reef emergence at Nakassa Island, but the island uplifted at least 0.85 m m in 2007 (Taylor et al., 2008). Coral morphology in 2012 was similar to that at Ghoi Island with uplifted microatolls having depressions in their centers that indicated rapid submergence and upgrowth upward and inward from their living peripheries over recent decades. Because the corals had grown upward a great deal, we elected to drill into one for its relative sea level record rather than try to saw a cross section. However, none of the microatolls was as large as at Ghoi so that the longer history of vertical motions is not preserved.

Microatoll NAK-A is about 3.5 m diameter and 1.5 m tall on sandy bottom. The seven vertical cores show the unconformity in coral growth that represents its HLS prior to submergence and upgrowth. The cores trace growth from the outer edge that is all younger than the unconformity inward as growth covered the interior floor of the microatoll. Core NAK-H exhibits the first part of the unconformity to be overgrown near the edge of the coral. Core NAK-A shows continuous upgrowth outboard of the unconformity. Cores G, B, E, D, and C show the progression of growth inward toward the center of the microatoll as growth covered the former HLS. Dates above and below the unconformity of 1960 ± 0.69 and 1956 ± 0.86 C.E. bracket the time when upgrowth from the living coral began at the periphery at about 1959 C.E. The coral grew upward a total of ~0.55 m at a mean rate of ~11.5 mm/yr.

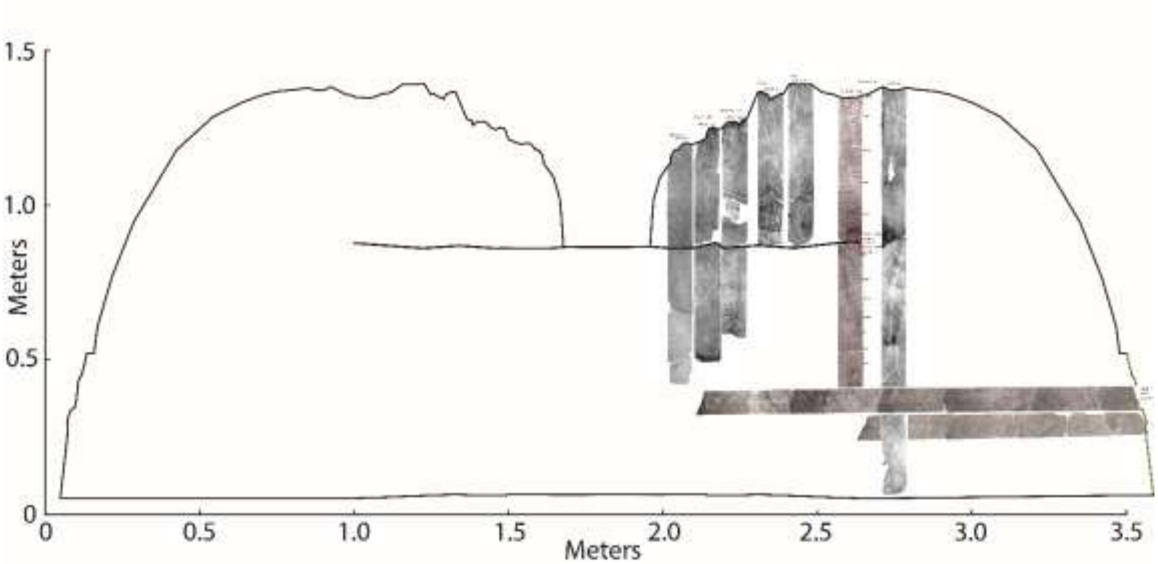
However, the four cores that penetrated beneath the unconformity clearly show that before the top of the Microatoll died in about 1957 C.E. that it had been growing straight upward at a mean rate of 11 mm/yr with no evidence of stress. We can assume that there was abundant accommodation space. However, the sudden death of the top of the coral occurred just about the time of the prolonged 1957-59 ENSO event accompanied by lowered sea level (SOI record) Becker et al., (2012) does not support a significant sea level anomaly at this time. However, the top of the coral did die and we assume that the tops of nearby microatolls with similar morphology died at about the same time. The living top of this coral in 2007 may have been some distance below its potential HLS when it uplifted so that the 0.85 m of coseismic emergence may be a minimum. The 0.85 m of emergence is ~0.3 m greater than the amount of upgrowth of the coral after its top died. Because Nakassa Island has no net Holocene uplift emergence, there should be similar amounts of tectonic subsidence and tectonic uplift although that balance does not have to be perfect over one cycle. However, it is possible that tectonic subsidence began before the coral top died and continued or resumed after the top died. It is possible that the dead top records a 1957-59 ENSO lowered sea level, but it is at least as likely that a temporary uplift emergence event occurred that killed the top of the coral.

One of the largest earthquakes in this part of the Western Province occurred on 17 August 1959 and is known to have caused some vertical displacement. The epicenter was located ~30 km northwest of the Northern end of Ranongga and almost due west of Mbava. A 5 km length of coastline near Supato Village on southern Vella Lavella subsided and the shoreline retreated ~60 ft in one month (Grover, 1965). No uplift was reported and most of the reported subsidence may have been due to compaction. However, the event had a USGS-reported Mw 7.0 and shallow depth (25 km), which is consistent with the depth to the interplate megathrust as estimated from coseismic uplift accompanying the April 2007 earthquake. There could well have been uplift during the 1959 event as well.

Houses were destroyed and fissures opened on both Ranongga and Vella. A tsunami was generated on the west coast of Ranongga and swept around the north end of Ranongga and onto the northern east coast. Most landslides were on Vella rather than Ranongga and shaking may have been more intense on Vella. Damage also on Gizo. Tsunami generated off the west coast of Ranongga. At Binskin Island, NW Mbava, apparent subsidence of 5 ft. (But this could have been compaction). Present low water mark is higher than previous high water mark (1959 interview of Mrs. Biskin). But not all of Biskin Island subsided 5 ft. Solid coral rock parts subsided only ~1ft.

Grover, J.C., 1965, Report No. 57: The Great earthquake in the Western Solomons on 18 August, 1959 17/2105 G.M.T. in The British Solomon Islands Geological Record, Vol. II – 1959-62, p.174 – 182.

It is entirely possible that tectonic uplift played a role in both the death of the top of this previously healthy microatoll as well as its recovery and rapid upgrowth from ~1960 until it was uplift ed in 2007.



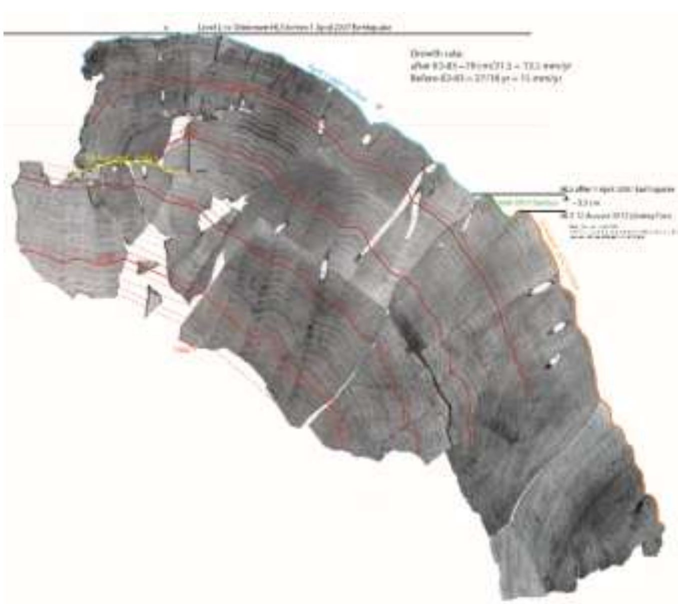
Mbava Island

The southern end of Mbava Island has emerged coral limestone to elevations of at least 6 m near the shoreline. A modern bioerosion notch is cut into cliffs in the limestone from near the modern mid-tide level up to at least 2.5 m higher. The cliff consists of pre-Holocene coral limestone and there is no unequivocal emerged Holocene coral limestone other than the tops of coral heads that emerged in association with the April 2007 earthquake. We concluded that there is little or no net Holocene uplift (Mann et al., 1998) other than ~0.5-0.7 m of 2007 coseismic uplift. That amount of 2007 emergence is a minimum because corals were actively growing upward and may not have kept up with rising relative sea level. The tall bioerosion notch, the absence of net Holocene uplift, and 2007 coseismic uplift suggest that elastic strain accumulation and release cause relative sea level to oscillate within the vertical range indicated by the bioerosion notch in the sea cliffs.

Many of these microatolls have depressions in their centers that record upgrowth from a previously established HLS. However, the depth of the 0.3 to 0.4 m depths of depressed centers is less than the amount off 2007 coseismic uplift and is less than the upgrowth of the the Nakassa and Ghoi microatolls.

Microatoll 12MB-A has a vertical dimension of 1.17 m with living coral from near the bottom up to 0.8 m higher.

A cross section of Porites microatoll 12MB-A shows that the coral had been growing upward at least since 1964 C.E. Unfortunately, we were able to sample to a depth of only ~55 cm. Its uppermost surface died in ~1983. This is the surface that forms the depressed dimple in the center of this microatoll. There also is evidence of stress and some dead surface area in 1993 C.E. The coral grew relatively fast and with clear annual banding both before and after the death of the top of the coral in 1983. From the top surface of the 1983 dead coral top it is evident that only the relatively flat top-most surface of the coral, but not even its uppermost sides, died and it immediately began to regrow upward and horizontally over that surface until the 2007 uplift.

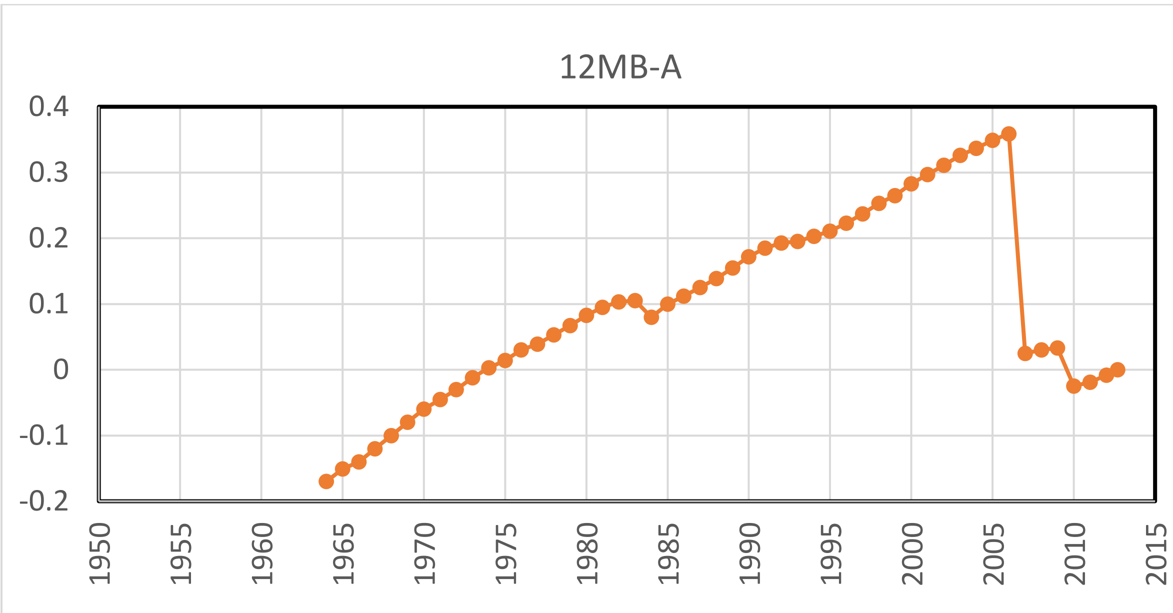


The upper 0.33 m of the coral was initially killed down by emergence in 2007. The uppermost surviving coral grew upward until years and then died down an additional 0.05 m in 2010. By late 2010 until collected in August 2012 the coral grew upward by 0.03 mand horizontally by a few cm. Total emergence of this coral appears to have been 0.37 m. However, most other corals indicate 0.5 to 0.68 m of emergence. This suggests that 12MB-A was living well below its potential HLS in 2007.

The group of Mbava Island microatolls appears to have colonized a shallow substrate at about the same time. Soon after growing upward about 0.3 m their tops were partially emerged. The consistent -.3-0.4 m depth of central depressions indicates that all coral tops died in about 1983 as did 12MB-A. Most were growing in water depths of only ~1 m.

The ability of the sea level low of 1983 to reach low enough to kill the tops of corals suggests they were then living near their HLS and that the 1983 ENSO sea level lowered the HLS enough to kill their tops.

However, they quickly recovered after 1983 and grew upward and inward over the 1983 dead surface. This upgrowth probably was facilitated by net sea level rise after ~1993, but there was surely a component of tectonic subsidence. Tectonic subsidence may have been less than farther south at Nakassa and Ghoi where corals grew upward by 0.5 to 1.0 m, but it still was an important factor and kept low ENSO sea levels in 1998 and others from killing the coral tops. Because many microatolls showed less 2007 emergence than others it appears that coral upgrowth may have been hindered by the 1983 killing of coral tops or the tectonic subsidence plus sea level rise exceeded the ability of many microatolls to keep up with net sea level rise. It is possible that tectonic subsidence after 1983 was faster than before. It also is possible that there was some tectonic upift of southern Mbava in 1959 that raised the corals so they were more vulnerable to emergence during the 1983 ENSO low sea level.



Slower growing Goniastrea spp. microatolls had grown upward much less after emergence last killed their tops, but they do show a similar sequence that includes emergence death of their tops, upgrowth, and then death by emergence in 2007 (Fig. x).

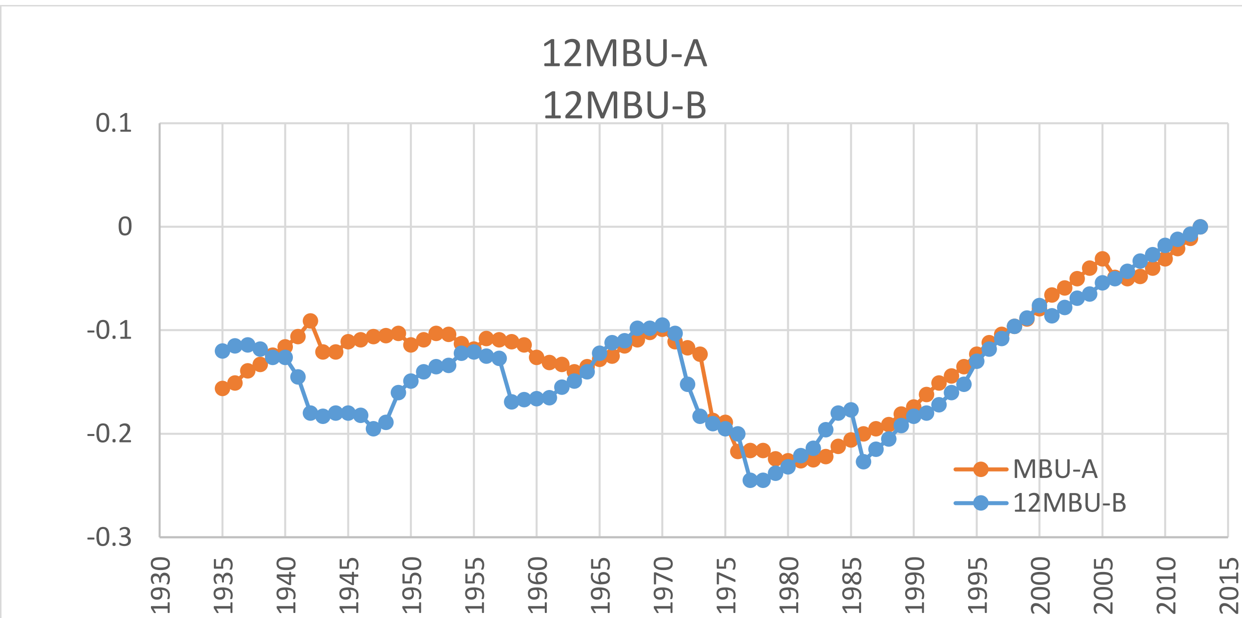
Northwestern Ghizo Island

The northwestern-most north coast of Ghizo Island is located very near the hingeline between coseismic uplift and subsidence during the April 2007 earthquake (Taylor et al., 2008). This part of Ghizo has no emerged Holocene coral limestone.

At a small barrier island we cut sections from two Porites microatolls, 12MBU-A and 12MBU-B, which were about 50 m apart in very shallow water so that both are relatively thin. Neither of these corals show any evidence of uplift associated with the April 2007 earthquake, which is consistent with the observations of Taylor et al. (2008). The oldest part of the corals that we recovered is about 1935 C.E. The corals grew horizontally until 1970 with relatively little evidence of relative sea level change. The apparent fall of relative sea level in about 1942 in 12MBU-B may be real in that 1940-1942 C.E. was one of the longest and strongest ENSO and SOI events on record. The fact that is is not seen in 12MBU-A may mean that the microenvironment was protective of that coral at that time. However, both corals indicate a relative sea level fall of about 0.15 m in about 1972 C.E. After mid-1980's C.E., both corals grew upward steadily with only a minor pause in 12MBU-A in 2005.

This ~1972event is not seen in other microatolls and is likely to be related to tectonic uplift.

The steady rise from ~1980 resulted in 0.1 m of upgrowth before the regional sea level rise that began in ~1993 C.E. There is no evidence of growth interruption at the time of the 2007 earthquake. If there was any vertical displacement it was not sufficient to raise the coral above its HLS. If subsidence occurred it is not detectable. However, the possible tectonic uplift in ~1972 suggests that this area does undergo some amount of vertical deformation related to the earthquake cycle.



Ghizo Island

Emerged coral limestone occurs on several islands including on Nusatupe Island near the airport terminal, but it is evident by the deteriorated condition of the fossil corals that it is pre-Holocene. Mann et al., (1998) dated an emerged coral (5340 ± 120 yr 2σ by ^{14}C uncorr.) from an emerged reef ~1 m AHLC on Logha Island, the next island west of Nusatupe, and only 800 m north of downtown Gizo. Pre-Holocene coral limestone also rises to at least ~5 m. A bioerosion notch is cut into this limestone with its base near present mean sea level to about +1.5 m with the notch apex at ~1 m above mean sea level.

These data show that at least this part of Ghizo has not undergone significant vertical displacement over the late Holocene. The emerged notch matches the elevation of the emerged reef only 3 km to the southeast. Relative sea level changes caused either by tectonic or regional sea-level variability have not exceeded the level of the emerged Holocene reef and notch for a duration sufficient to erode a higher notch or to grow a reef. Because we do find some 2007 coseismic uplift along the south coast of Ghizo and the absence of 2007 uplift, and more likely subsidence, in the microatolls at 12MBU-A and B, it appears that Ghizo has oscillated vertically by small amounts in association with earthquake cycles during the late Holocene.

South of Olasana

12SEG-A is a partially emerged Porites microatoll about 8 m in diameter in water about 1 m deep and is located within the 2007 coseismic uplift zone. Corals living a few hundred meters south uplifted ~0.3 m in 2007 (Taylor et al., 2008). There is no evidence of any net uplift of any net Holocene uplift of this area, which is located only 1-2 km south of the 2007 hingeline.

The central area of this microatoll stands about 0.5 m higher than the wide lower area of more recent coral growth. This lower terrace of coral growth is symmetrical around the emerged center and was still living on its outer edge in 2012. Based on the ~2 m distance from the emerged central area to the living coral and a 11 mm/yr growth rate, a rough estimate is that the lower coral ring grew for 182 yr after it emerged or since 1830 C.E.

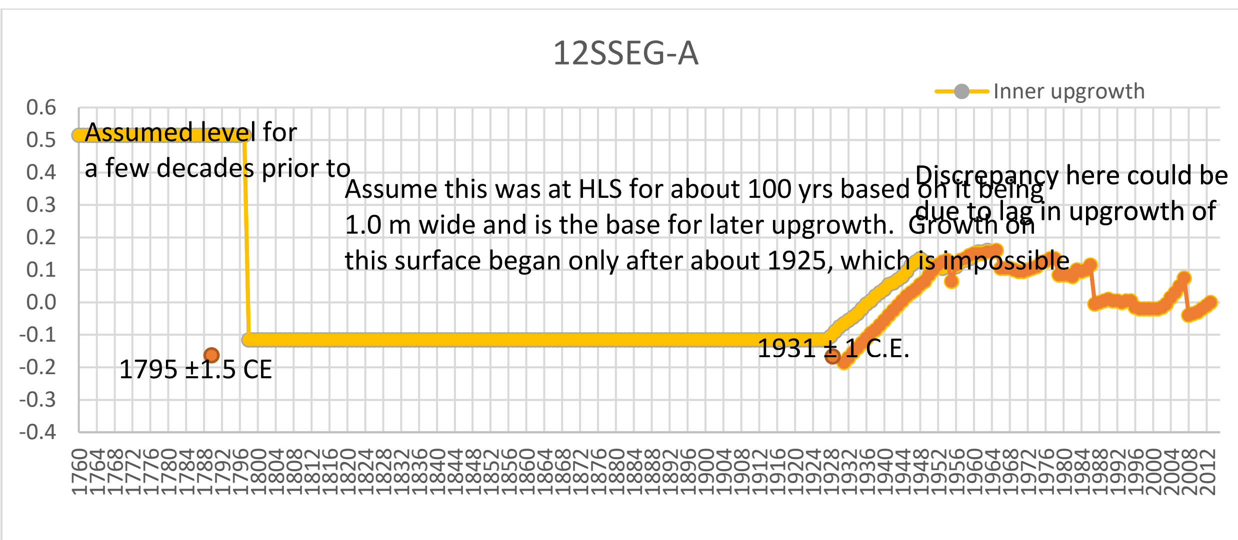
Although we slabbed the coral as deeply as possible, we did not sample the first coral that grew after this central emerged coral. Instead, the from the emerged part outward for the first meter we sampled coral overgrowths that grew on top of the underlying coral that grew out horizontally from the emerged central part of the microatoll, but at a level about 0.4 m deeper. The slab broke off at the boundary between the deeper coral and the two coral overgrowths. However, the outer part of the slab shows that the coral had grown upward and outward from a location underlying two coral overgrowths.

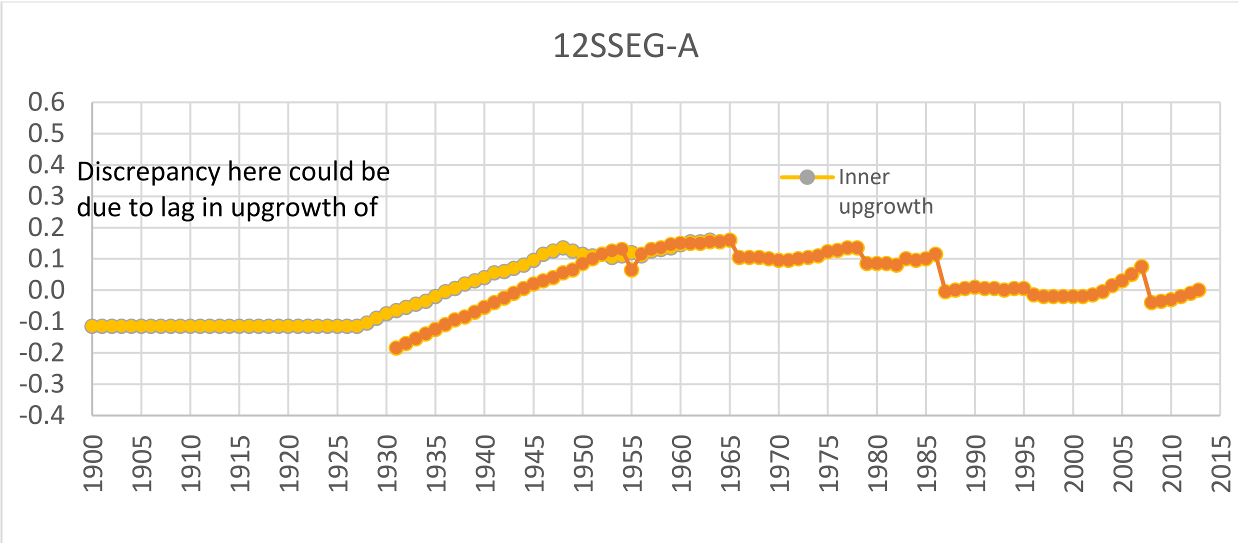
The coral morphology indicates that the coral emerged roughly ~1830 C.E. by about one meter. It then grew horizontally at a level at least 0.4 m deeper than the subsequent overgrowths. After growing horizontally at least one meter, relative sea level began to rise. The coral grew upward and outward and was living on its top and its periphery when it was uplifted ~0.3 m in 2007. In 2012, only the outer periphery was still living. The coral did not grow inward over the older dead surface because the two centers of coral overgrowth occupied that space. By counting annual growth bands and using two ^{230}Th dates we have constrained the sequence of relative sea level changes.

The outer edge of the high-standing central area gave a ^{230}Th age of 1795 ± 1.8 C.E. This is a maxiumum age for emergence. It is a few cm within the emerged coral and there may have been perhaps 0.1 m of erosion of that surface if not more before it was covered by subsequent coral growth in the mid-20th century. Another date near the origin of the inner overgrowth (which covers the lower surface of the older central area killed by emergence) indicates that this overgrowth began a little before 1931 ± 1 C.E. Band counting indicates that the second overgrowth began at very nearly the same date, but from a location ~0.1 m deeper. The bottoms of the coral overgrowths are not at the same depth which suggests that minor relative sea level variability caused some topography on the underlying coral surface as it grew outward. Coral growth bands counted inward from the living surface also indicate that the coral began to grow upward by ~1930 C.E., which is the oldest growth band recovered due to the depth limits of our saw.

We think that the coral grew horizontally at a depth just below reach of our saw based on the overgrowth of subsequent coral from that deeper surface.

Inner head grew up by 1931 ± 1 C.E.





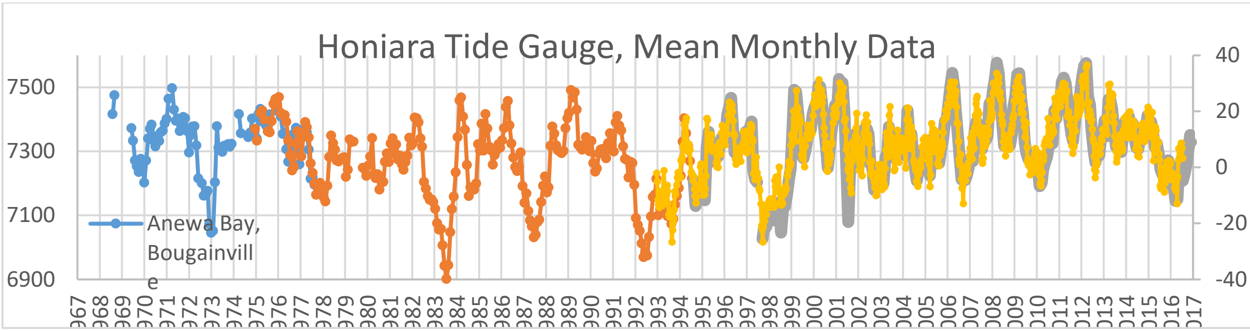
The emerged central area of this coral strongly suggests a paleo-uplift event sometime after 1795 C.E., and perhaps as late as 1830 C.E. This uplift was much larger than the 2007 uplift of only ~0.3 m and appears to have persisted until about 1925-1930 C.E. About that time, the coral began to slowly subside and grew outward and upward by ~0.4 m and reaching its highest level by ~1965 C.E. Erosion may have removed a few cm of additional upgrowth. In ~1965 C.E., the coral died down by ~0.1 m and grew horizontally and upward by ~0.05 m from ~1972 until 1978. It then died down again by ~0.05 m and grew horizontally until ~1987 when it died down an additional 0.12 m. This diedown was possibly related to sea level lows during the 1983 and 1987 ENSO events because, rather than growing back upward as it should following a temporary low sea level, coral growth maintained this level until 2001. Then the coral abruptly began to grow upward very rapidly until coseismic uplift in 2007 killed the upper 0.15 m of living coral. Other nearby corals indicated that the uplift was ~0.3 m, which suggests that 12SEG-A coral growth was lagging behind relative sea level rise and that there may have been a tectonic subsidence component to the relative sea level rise. Rapid coral upgrowth resumed immediately following 2007 emergence. This upgrowth could record tectonic subsidence, but sea level was also rising so that we cannot currently distinguish the causes.

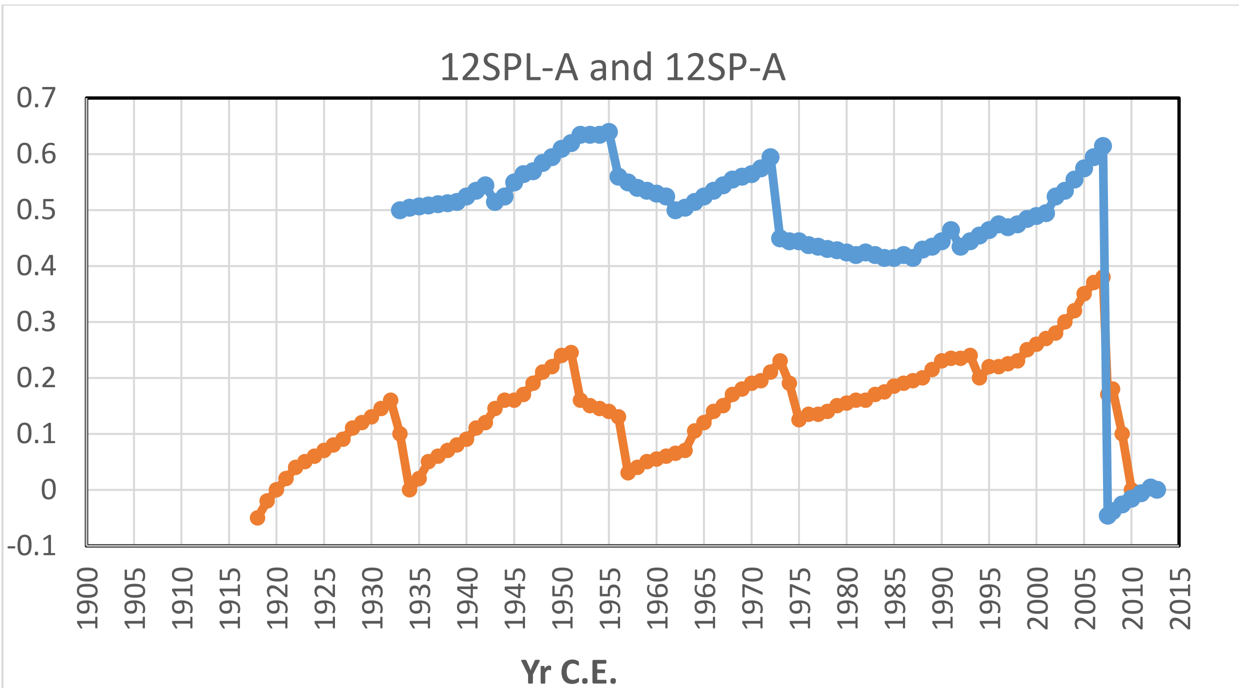
12SPL-A and 12SP-A

T he southern coast of Parara Island uplifted during the 2007 earthquake and many microatolls along the southern barrier reefs of western Parara Island and were partially emerged during the 2007 earthquake. None of these barrier islands have emerged Holocene coral limestone, but Parara has coral limestone reaching to tens of meters above sea level. Bioerosion notches in these cliffs indicate that total Holocene emergence at this location is no more than about 3 m. We have found no higher notches or evidence of higher Holocene coral on Parara. Mid-Holocene sea level was approximately 2 m higher than present sea level and declined due to GIA. That fall of sea level could account for emergence of the upper notch. The lower notch was occupied by sea level prior to the 2007 earthquake, but its vertical extent is larger than typical. Note the white line of coralline algae at the base of the lower notch that had bleached due to emergence one month after the earthquake (Fig. x). This may indicate that vertical tectonic movements have caused relative sea level to oscillate within the vertical extent of this notch, and possibly the higher notch as well.

We sampled two microatolls on the barrier reefs with 12SPL-A located ~6.5 km northwest of the notched sea cliffs on Parara. Microatoll 12SP-A is located another 2.5 km west of 12SPL-A. Both are about 4 m in diameter and have very similar surface topography, Nearby microatolls appear very similar. 12SPL-A was still living when sampled and had emerged.

Microatoll 12SP-A has a longer record that shows ~0.2 m of upgrowth from ~1917 until ~1933. However, this upgrowth may simply be the coral growing up to its HLS from a deeper origin. However, after ~1935 C.E., both corals grew upward until 1950 and 1955 C.E. The discrepancy in this upgrowth could be due to errors in the growth band chronlology. 12SPL-A grew upward ~0.15 m while 12SP-A grew upward ~0.25 m. Both corals then died down about the amount they had grown up, one in ~1955 and the other in ~1951. Again, the difference could be a growth band counting error.Both corals began growing upward again From ~1960 until1972-73 C.E. when they and almost reached their previous levels after which both again died down by 0.12 to 0.15 m. 12SP-A began growing upward immediately while 12SPL-A died doawn an additional 0.05 m over the next 15 yr until it, too, began growing upward in ~1987 C.E. By 2007, both corals had grown back up to or slightly above their previous two peak levels. The 2007 earthquake caused 0.65 m and 0.27 m of emergence of 12SPL-A and 12SP-A, respectively. However 12SP-A coral surface died down an additional 0.17 m for a total of 0.44m of apparent coral emergence by 2010 C.E., after which it died completely.





In addition to the large emergence events affecting these corals near the time of the 2007 earthquake, both of these microatolls indicate two significant cycles of apparent submergence that ended with their abrupt emergence. A final period of submergence of similar amplitude preceded the 2007 earthquake. These relatlive sea level changes do not correspond to any regional sea level changes from the Honiara tide gauge except for the submergence preceding the 2007 earthquake. That submergence caused ~0.26 m of upgrowth in 12SP-A, but it began in 1975, well before the 1993 onset of accelerated regional sea level rise. After 1993, 12SP-A grew upward only~0.15 m. Most of the 0.2 upgrowth of 12SPL-A occurred from ~1987 until 2007 which does correspond pretty well with the 1993 onward sea level rise. Both coral upgrowths appear to have accelerated in ~2002. It is likely that both corals were at least 0.1 m deeper than their potential HLS's as they grew upward and were unable to keep pace with relative sea level rise. Otherwise we would expect that ENSO-lowered sea level in 1998 would have killed the coral tops.

We interpret the four coral diedowns in ~1932, 1951-1955, 1972-1973, and 2007 to represent abrupt tectonic uplifts. We can find no evidence of earthquakes of sufficient magnitude to possibly account for any but the 2007 event. We further interpret the submergence of the corals preceding each emergence event to represent either gradual or abrupt subsidences. We are not able to distinguish the durations of the subsidence events, but only that they created accommodation space allowing coral upgrowth to proceed at its maximum possible rates.

Lola Island

Parara and Kohinggo Islands on the north and south sides of the Vona Vona Lagoon have uplifted in late Quaternary time and we assume they also subsided afterward until uplift began sometime probably before ~20 ka (Taylor et al., 2005). Evidence of 2007 tectonic subsidence at the time of the 2007 earthquake in the Vona Vona Lagoon is very clear, including at Lola Island where the grounds were still being flooded during high tides three weeks after the 1 April earthquake. We estimated subsidence of 0.51 m during our visit based on the reach of high tides. However, by the next visits in 2009 and 2010, only the king tides inundated the hotel yard (hotel owner Joe Entriken confirmed that the island had gone back up to near its pre-2007 level). Taylor et al. (2008) presented only the evidence of subsidence, but the uplift appears to have been very nearly the same amount and no net vertical displacement is evident.

New Georgia

Near Munda Airport

All of New Georgia including the barrier islands of the Roviana Lagoon, the Diamond Narrows leading up to Noro, Kohinggo Island, and Kolombanggara Island have uplifted at least since mid-Holocene time (Mann et al., 198; Taylor et al., 2005). Some uplifted coral that was not re-submerged by the late Pleistocene subsidence remains around Munda and the barrier islands to the south. Holocene reef uplift rates range from ~0.5 to 1.2 mm/yr. There appears to have been at least one uplift along the New Georgia coast from the Munda area to well north of Noro Port within the past 1000 yr. The full extent of this uplift and whether it was abrupt and related to megathrust ruptures is unknown. We recognize it as an emerged corals fringing the higher Holocene reef. One sample from 1.65 m ALC from this emerged reef north of Noro gave an age of 738.6 ± 3.0 or 1273 ± 3.0 C.E.

On 24 April 2007 at Agnes Lodge on the shore ~300 m south of the airport terminal the water stood on the yards after high tides (Fig. x). The entire shoreline in and around the area clearly subsided. However, as at Lola Island, when we returned in 2009, 2010, and 2012 we found that the shoreline had re-emerged and much or most of the subsidence had been erased. This observation is confirmed by GPS measurements of a campaign station at the Munda airport that showed 0.3 m of subsidence from 2001 to May 2007 and 0.x m of uplift from May 2007 until 201x. About 300 m offshore small Porites microatolls had dead tops, but had begun to grow upward on their living margins since approximately 1997 (Fig,. x). We have not re-measured offshore microatolls to assess the exact amount of uplift. The GPS site at the airport was destroyed during a 2013 re-build, but we do have campaign data from the airport GPS site for 2009 and 2010 that can be processed.

Near the Munda Inlet ~5 km southwest of the western end of the airport we examined microatolls in water that was still very murkey from the 2007 earthquake. Some microatolls were 2-3 m diameter and had dead tops with about 0.1 m of ongoing upgrowth at their margins (Fig. x). Their tops previously had been killed by a lower sea level that we estimate to have occurred ~1997. Relative sea level then rose and the microatolls grew upward. We estimated ~0.6 m of subsidence in 2007 at this location (Taylor et al., 2008) which was confirmed by Saunders et al. (2015) who investigated proliferation of coral growth in 2011 and 2013 to accommodation space created by subsidence in 2007. However, they assumed that mean coral growth of 0.17 m began in 2007 and was 28 mm/yr. Unknown t them, we found upgrowth of ~0.1 m had already occurred in April 2007 so that the growth began perhaps a decade before 2007 and was therefore closer to 10-11 mm/yr than 28 mm/yr (Fig. x). Eleven mm/yr to be a typical extension rate for Porites microatolls in the Western Solomons.

Microatolls along the western New Georgia coast north of Noro also have dead tops with margins that began upgrowth even before the 2007 subsidence. We found no living microatolls larger than ~2 m diameter.

The barrier islands of the Roviana Lagoon near Munda had subsided in 2007. We observed this subsidence until Sasavele Pass, but at xxx pass there was no evidence of 2007 subsidence and microatolls near the pass indicated emergence over the past decade rather than submergence. The emergence was slight, but clear. Microatolls grew on a dead surface that appeared to consist of larger Porites corals. We observed no large microatolls.

Summary for the Munda area. The whole area appears to have very few microatlls more than a few m diameter although there are dead emergent microatolls at the shorelines that are many meters in diameter. Oddly, we found no large microatolls in shallow water, as if they had all died a few decades ago. However, a ~4 m diameter Porites sp. head coral was seen in deeper water growing about 1 m below the 2007 HLS and thus ~1.5 m deeper than corals could grow in 2012. Living microatolls appear to be rooted on a dead coral surface from which they grew upward until their HLS. Those in the 2007 subsidence zone had tops killed in approximately 1997, which is about the time of the 1997-98 ENSO low sea level. However, these corals probably had to have undergone some uplift as well as upgrowth to be affected by the 1998 ENSO low sea level because their HLS's would have been trimmed by the much lower 1983 ENSO low sea level. The dead tops of corals toward the eastern limit of the Roviana Lagoon suggests that the entire area may have undergone relative emergence at about the same time in the late 1990's although those near Munda then underwent submergencae upgrowth while those near eastern Roviana underwent additional emergence.

If we assume that after death that it was soon colonized by Porites corals, then the sizes of the living corals suggest that the dead corals must have died around the middle of the 20th century. The domed microatoll dead tops could mean that the coral tops were below their HLS and growing upward when they were exposed and killed during an ENSO low sea level as late as 1998. This is consistent with rising sea level that began about 1993 (e.g., Beck et al., 2012; Tide gauge data). After their tops were killed they would have had until 2007 to grow their outer living periphery. This is about the right scale of relative sea level change and coral growth.

2013 photos by Megan Saunders

Rendova

All of Rendova has undergone Holocene emergence except the northernmost coast and barrier islands. The island has tilted down to the north so that Holocene uplift decreases to zero near the north coast. Across the Blanche Channel on the Roviana Lagoon barrier islands, Holocene uplift of again at rates of ~1 mm/yr (Mann et al., 1998). Corals and GPS campaign site data show that Rendova underwent very little net vertical deformation during the 2007 earthquake (Eric Kendrick, personal communication, 2008). However, at least parts of the southern coast of Rendova underwent about 0.5 m of subsidence during the January 2010 earthquake between the shoreline and the trench (Newman et al., 2011 and our observations). In August 2012 we sampled two microatolls near Ughele Village on northeastern Rendova and made microatoll observations in the Rava Point area.

Ughele Lagoon South

The Ughele area has uplifted on the order of 6-8 m since initial mid-Holocene emergence (Mann et al., 1998). The most recent uplift of this area is best seen near Husuzu where total emergence since the mid-Holocene is ~20 m and in situ corals (99RND-J and 99RDN-K) at 2.2 m ALC gave ages of 402.2± 2.1 and 404.1± 2.0 yr. This uplift event is seen in sea cliffs at steadily decreasing elevations northward along the coast until it is indistinguishable near Ughele due to erosion and decreasing vertical offset from living coral and intertidal bioerosion. We have detected no more recent vertical displacement in this area except for whatever we find in the microatolls.

Located just south of Ughele village, microatolls in this small lagoon with a large entrance have a relatively simple morphology. The chronology of this coral was confirmed with a ^{230}Th age of $1947 \pm 12\ yrs$. The relatively large error in this age and its disagreement with the fairly good annual banding in the coral leads us to rely strongly on the growth band chronology. However, because the coral was living when sampled, the chronology is not likely to be in error by more than a few years.

12UGH-A grew horizontally from the time it grew up HLS in ~1933 CE until 1968 when it appears to have abruptly emerged by ~0.1 m. Horizontal growth without significant relative sea level changes persisted until ~1978 when relative sea level began to rise until pausing from 1980 until 1987. It resumed rising, with only a small diedown of 0.03 m in 1991, until we collected it in August 2012. The total 1978 to 2012 upgrowth was 0.3 m. After 1991 the coral was growing at ~13 mm/yr with no evidence of disruption by ENSO-lowered sea levels or reaching near its HLS. This suggests that the relative sea level rise could have been at least a few cm greater than the net upgrowth of 0.3 m. There also is no evidence of any effect by the 2007 or 2010 earthquakes which is consistent with the conclusion of 2010 subsidence (Neuman et al., 2011).

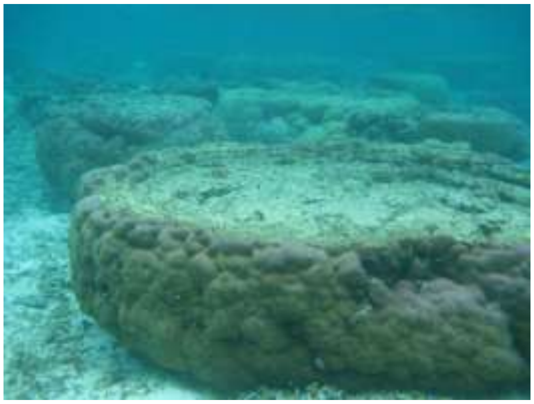
From microatoll 12UGH-A we interpret a possible small uplift of ~0.1 m in ~1968 that is not explained by known sea level change. The ~1968 emergence was abrupt and sustained until 1978. Upgrowth from 1978 accelerated after 1991and reached a total of 0.3 m by mid-2012. Regionally rising sea level may have contributed to ~0.1-0.15 m of upgrowth after 1991, but the remaining 0.1-0.2 m of submergence is likely to be caused by tectonic subsidence of at least 0.2 m given the uninhibited ongoing upgrowth until 2012. This subsidence is roughly consistent in timing and amount with that seen in microatolls farther west along either side of the 2007 hingeline. It is possible that the northern Rendova area also subsided somewhat in 2010 if the large subsidence of the south coast and eastern Tetepare extended this far north (Neumann et al., 2011)

Rava Point

The Rava Point area has coral limestone emerged to ~4 m ALC indicating only very slight net Holocene uplift (Mann et al., 1998). GPS measurements from 1997 until May 2007 (Kendrick, personal communication 2008; Wallace et al., 2014) show that Rava Point subsided ~13 cm in 2007 according to a nearby campaign GPS site. Rava Point also subsided by ~0.5 m in the 2010 earthquake (Neuman et al., 2011) which killed coastal trees and shrubs and left some trees standing in the sea. (Fig. x)

On 11 August 2010 we observed numerous microatolls in the nearby shallow lagoon up to 2 m in diameter. Note the distinct difference from 12UGH-A. All had grown up a few cm beginning in the preceding few years. Slight subsidence in 2007 may have contributed to upgrowth by August 2010. However, subsidence in 2010 was too recent to have affected the corals. Their centers are 5-10 cm lower than their living peripheries which may be a result of longer-term net sea level rise. Each had undergone repeated small upgrowths and diedowns on the order of only a few cm each over the past 50 to 75 years. There clearly were no dramatic changes in relative sea level. There is no evidence of the 0.1 m emergence of 12UGH-A in ~1968 or the relative sea level changes seen in corals near Munda, Parara, Gizo and Ranongga. Indeed, there is scant evidence of response to the ongoing regional sea level rise of ~10-15 cm (e.g., Becker et al., 2012). A possible explanation is that this part of Rendova was uplifting enough to cancel the regional sea level rise until the 2007 and 2010 seismotectonic subsidences.

The apparent net upgrowth over the past ~100 yr of coral growth could be attributed to some net regional sea level rise over this period, consistent with global trends. The slight up and down growth patterns are typical of the effects of a series of ENSO-induced low sea levels on microatolls. However, regional sea level rise should have caused the coral to have grown up by at least 10 cm since ~1993. The fact that this did not happen suggests that tectonic uplift may have been compensating for regional sea level rise, possibly related to elastic strain accumulation in advance of the 2010 coseismic subsidence (Neuman et al., 2011).



Tetepare

Tetepare has the highest known Holocene uplift rates in the Western Solomons with a mid-Holocene reef up to at least 47 m ALC along its mid-northern coast (Mann et al., 1998; Taylor et al., 2005). However, the small western peninsula where the TDA facilities are located uplifts at a much slower rate with Holocene reefs reaching to only ~10 m ALC. In August 2012, we observed living microatolls at two well-separated locations on Tetepare. These are: 1. At the southern side of the westernmost small peninsula of Tetepare where the Holocene reef is much less emerged , and 2. Along the central southern coast about 7 to 18 km east from the western end where the Holocene reef reaches to at least 47 m ALC.

The Holocene coral reefs provide context in which to view the microatolls and modern vertical motions. Tetepare has Holocene reef at least as high as 47.5 m ALC (Mann et al., 1998) along its central northern coast inland of the 12RAR-A and 12TET-A microatolls described below. The mean late Holocene uplift rate is then 7-8 mm/yr in this area. However, Holocene reef on the small western peninsula of Tetepare reaches to only 9.25 m ALC where a coral gave a date of 7636 ±32 yr. The mean uplift rate here is then about 1 mm/yr and much slower than farther east. The western peninsula must be downdropped relative to the rest of Tetepare. This provides context for the ages of the morphologically intact in situ fossil corals 12NWT-A and 12NWT-B at the shoreline on the north side of the peninsula 1.3 km from the west end. These corals are on a subterrace level at 1.2 m ALC and gave ages of 1173 ±6.7 and 1442.6 ± 4.7 yr. This is the most recent emergence of western Tetepare that was large enough that coral branching morphology could be preserved. Farther west, at 3 km from the west end on the main body of Tetepare, 12BT-A and 12BT-B are in situ encrusting corals at 4 m ALC where the top of the Holocene reef is at least twice the elevation at at the previous site. This coral gave a ^{230}Th age of 2232 ± 7.5 yr. No other well preserved coral was found between this coral and sea level. Farther east a number of coral samples from just above sea level were previously dated for paleoclimate studies. The youngest was 3089 ±25 yr. Thus we found no evidence of well preserved corals that indicate paleouplifts large enough to have raised corals above intertidal bioerosion during the past two thousand years on central northern Tetepare and the youngest corals become progressively older as the mean uplift rate becomes progressively greater. It appears that large coseismic uplifts have not affected the main body of Tetepare for at least 2000 yrs and affected western Tetepare only as recently as 1200-1400 yr ago.

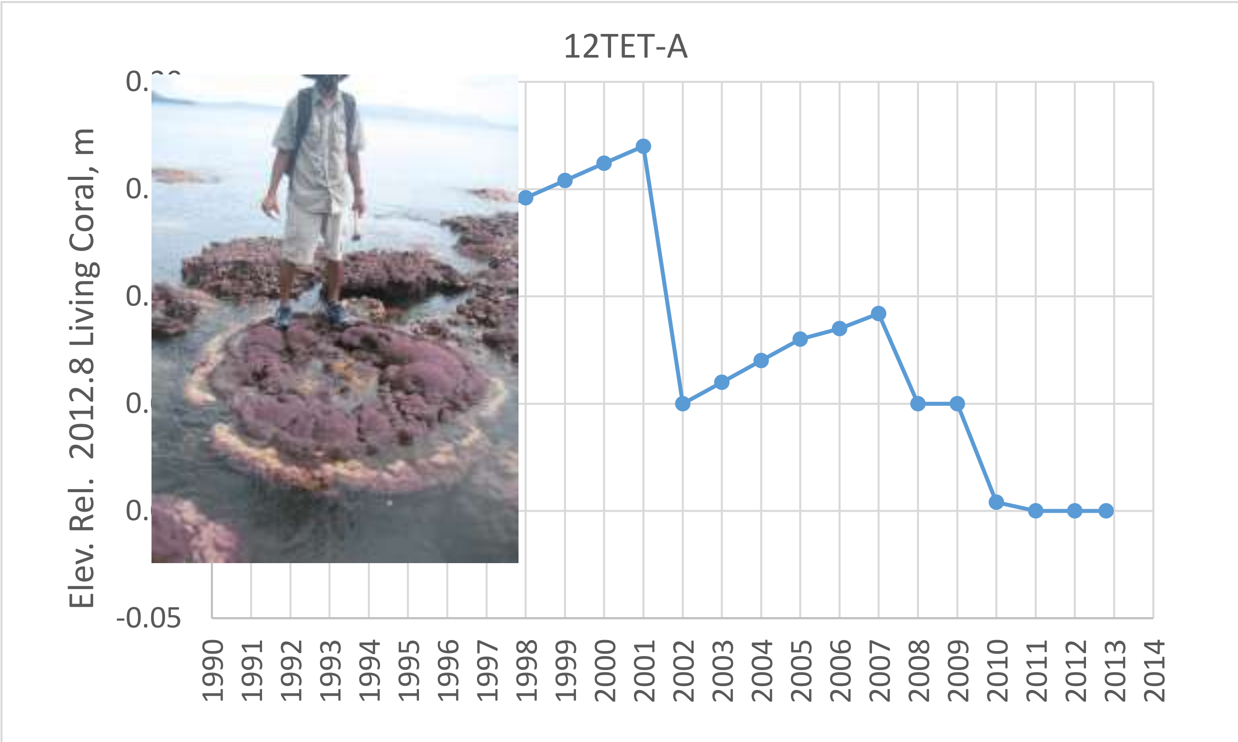
As for modern relative sea level changes, in April 2007 we visited the TDA lodge and found evidence of at least a 10-20 cm of recent subsidence in addition to the opinion of witnesses that the area had subsided during the earthquake (Taylor et al., 2008). However, we had no quantitative measurements to support this assertion.

In August 2010 and August 2012, we found at the TDA lodge area on westernmost Tetepare clear subsidence on the order of ~0.5 m in the form of shoreline transgression into the coastal forest, dying trees standing in sea water, and submergence of the boat landing. Witnesses assured us that the subsidence occurred at the time of the earthquake in 2010. In August 2012 we also found a cluster of Porites microatolls up to 3 m in diameter a few hundred m west of the boat landing in about 0.5 m of water at low tide. Other smaller mircroatolls occurred scattered over a few hundred m to the west. The outer rims of all microatolls had grown upward and inward over their dead tops by 6-8 cm in August 2012. This is too much growth to have started in only 2010, but is completely consistent with upgrowth starting in 2007 at a rate of ~ 14 mm/yr. This contrasts with the microatolls just west across at Rava Point, Rendova, that had not even begun to grow upward when we found them in mid-2010. It appears that the western end of Tetepare subsided in 2007 even though it is farther from the 2007 rupture zone than Rava Point, which did not subside. However, both the Rava Point and TDA microatolls share the characteristic that neither display any evidence of relative sea level changes before they subsided. This is surprising given that sea level had risen by 10 - 15 cm from 1993 to ~2012. This situation strongly suggests that most of the regional sea level rise on western-most Tetepare was cancelled by tectonic uplift until the subsidences associated with the 2007 and 2010 earthquakes.

From the west end of Tetepare until 7 km to the east we found no evidence of coral emergence. However, beginning at 7 km we began to find coral emergence on the order of 10-15 cm. At the farthest point on our trip at a small island at 18 km east along the south coast where corals showed recent emergence of 20-25 cm.

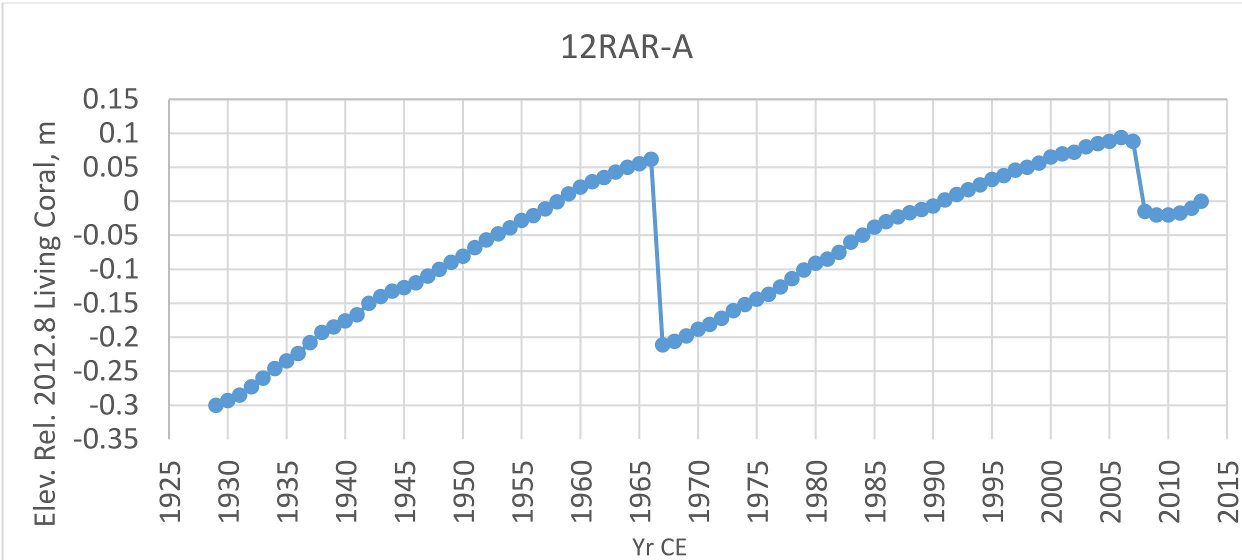
We sampled a small living Porites coral in June 2012 in shallow water at about 10 km from the west end of Tetepare, and sawed a larger one about 11 km from the west end (Table x). The smaller coral emerged by about 0.12 m in 2001, ~0.05 m in 2007 and another 0.05 m in 2010.

Although the 2010 emergence may have been related to the 2010 earthquake, it is likely to be a delayed diedown from the 2007 uplift. The 2001 diedown is more problematic. It is larger than the 2007 emergence. There was a significant ENSO sea level low in 2001 that could have affected a coral that had grown upward to the maximum HLS by early 2001 and this coral grew back upward .044 m by 2007. This is most easily accounted for by a temporary climate-driven emergence event, but it also could have been a slow slip event that preceded the 2007 and 2010 earthquakes. Coral emergence occurred during a time of regional sea level rise on the order of 10 cm which increases the likelihood that both the 2001 and 2007 emergences were tectonic.



About one km farther west are numerous small microatolls that typically indicate a period of submergence followed by a fairly recent emergence event. One typical sample, 12RAR-A, underwent about 0.42 m of upgrowth from its oldest part in 1928 until 1966. This may have been normal upgrowth of a coral colonizing water deeper than its HLS. It also could signify relative sea level rise. In either case, in 1966 it emerged abruptly by almost as much as it had grown upward in the previous 38 yrs. It then immediately began growing upward again until by 2006 it reached nearly the same level as in 1966. Sometime in 2005-2006 the very topmost part of the head died down by about 1 cm. This indicates that they had grown closer to their potential HLS than had 12RAR. A strong ENSO sea level lowering in 1966 may have helped in producing this coral emergence. In 2007 the coral emerged 0.103 m and continued growing outward and slightly upward by a total of 0.025 m by August 2012. There is no evidence of the 2001 emergence event seen in sample 12TET-A only 1 km to the east. However, this general sequence of vertical motions is confirmed by the morphology of many of the nearby microatolls and the emergence in 2007 is clear from other living corals both here and farther east where emergence reaches 15-20 cm on living corals.

Other nearby corals that were not sampled, show a much larger emergence event at about 2000. (i.e., a couple of years before the main 2007 emergence that we attribute to coseismic uplift. (use at least one photograph).



Emerged fossil corals from Tetepare show that at least part of the main landmass of Tetepare has not undergone a major uplift over at least the past 2200 yrs and possibly 3000 yrs. We found no in situ corals less than at least 2000 years. However, two in situ fossil corals indicate that the western peninsula underwent at least 1.2 m of uplift 1200-1450 yr ago, but subsided during the 2007 and 2010 earthquakes. The western end of the main landmass uplifted in ~3089 yrs ago, but we found no evidence of this uplift farther east. In contrast to the subsidence of westernmost microatolls, 12RAR-A and other microatolls from the central north coast indicate uplift of ~38 cm in 1966. This was followed by either sudden or gradual subsidence that allowed coral upgrowth that, combined with sea level rise, removed all of the 1966 uplift. Emergence of a few cm occurred several years before 2007. We do not know if this pre-2007 emergence was tectonic. The 2007 uplift of 10 to 20 cm along the central north coast was smaller than the 1966 emergence, which we also attribute to tectonic uplift.

Both coral records show the 2007 and 2005 emergence events. 12RAR barely emerged in 2005 because it was too deep below its HLS to emerge the full amount. 12 Tet does record total emergence in 2005 and also emerged even more in ~1998. All of these emergence events are seen in other nearby corals (Fig. X).

The 2007 emergence is almost certainly related to slip along faults beneath Tetepare associated with the 1 April 2007 earthquake. Some large aftershocks occurred beneath Tetepare after the main event which did not itself rupture this far east. The 2005 and 1998 events also are likely to have been tectonic uplift. No significant nearby earthquakes are known for those years. The 1998 emergence could have been exacerbated by ENSO-lowered sea level at that time, but nearly every other microatoll was not affected in 1998.